

# Francisco D. Serna

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## Education

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**University of Minnesota - Twin Cities**

*Bachelor of Engineering in Mechanical Engineering*

**Expected May 2028 | Minneapolis, MN**

**GPA: 3.46/4.0**

## Internship Experience

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**Abbott Laboratories | *Electrophysiology Research and Development Intern***

**May 2026 - Aug. 2026 | Plymouth, MN**

- Led an evaluation study to validate polyurethane stopcock tubing as a replacement for existing PVC tubing in an introducing catheter by assembling test samples and executing **Design Verification (DV)** methods, including aging, water/air leak, tensile strength, and bend/kink radius testing.
- Collaborated in the development of a new test method using **Gage R&R** to characterize the variability of nitinol baskets through torque testing and evaluations of lateral stability
- Evaluated shaft inner diameter (ID) design changes through friction testing, development of **Design Control Models (DCMs)**, and **tolerance stack-ups** to assess the impact of increased ID on shaft interactions within the handle assembly using **MiniTab**, **SolidWorks**, and **Excel**

**Cretex Medical - rms Company | *Manufacturing Engineering Intern***

**May 2025 - Dec. 2025 | Coon Rapids, MN**

- Created a visual database based on research on the different types of tumbling media, cycle times, and their impact on post-tumbling properties, helping operators achieve a desired finish, saving up to 15 minutes between tumbling cycles
- Modeled and designed fixtures specific to individual components for laser marking and EDM wiring in **SolidWorks**
- Conducted inspections and calibrations of component iterations and machines weekly

## Work Experience

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**Lifetime Fitness Inc. | *Facility Operations Supervisor***

**Jan. 2023 - Present | Coon Rapids, MN**

- Directed club upkeep by overseeing 48 team members, ensuring the practice of standard club procedures
- Exceeded standard customer service values with existing and new members to ensure a friendly, clean environment
- Ensured team flow by completing scheduling, task lists, area designations, and other administrative tasks.

## Projects & Skills

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### **IOT Smart Trash Can**

- Designed and developed an IoT-enabled smart trash can using a Particle Photon 2, integrating ultrasonic sensing, load cell weight measurement, and servo-actuated mechanisms in **C++**, with real-time status indication via cloud
- Implemented cloud-connected monitoring and control through the Particle Cloud API, enabling remote operation and automated actuation, and modeled a 3D-printed enclosure in **SolidWorks**

### **Autonomous Spider-Bot**

- Designed and built a 12-servo hexapod-style robot in C++ on an Arduino Uno, engineering a custom pin/slider mechanism for coordinated leg rotation that lets the robot grip and pull itself across a surface
- Modeled the full mechanical assembly in **Creo** and **SolidWorks** and **3D-printed** all structural parts, integrating an IR proximity sensor to trigger an autonomous 90° turn when obstructions come within 25 cm

### **Technical Skills:**

- SolidWorks, C++, Minitab, MicroVu, PTC Windchill Workgroup Manager, PTC Creo, PTC Creo View, ANSI Drawing Standards, Sheetmetal design, Microsoft 365, Google Workspace, Bilingual in Spanish

### **College Activities/Societies:**

- President's Emerging Scholars Program (PES)
- Peer Mentor for first- and second-year students in PES
- American Society of Mechanical Engineers (ASME)
- Teaching Assistant for EE 1301: Intro to C/C++